MULHIM DELTA

Bioinformatic Analysis of Skeletal Muscle Gene Expression

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1. Dataset

GSE111551 – Blood and skeletal muscle gene signatures of exercise training in men (Muscle dataset)

2. Dataset Description

The dataset contains skeletal muscle gene expression profiles collected from 26 healthy, untrained male participants before and after an 18-week endurance running training program.

The study investigates how endurance exercise affects skeletal muscle gene expression and the biological adaptations associated with aerobic exercise.

3. Research Questions and Statistical Analysis

- Research Question 1

Does 18 weeks of endurance training significantly change skeletal muscle gene expression?

- Null Hypothesis (H_0)

There is no significant difference in skeletal muscle gene expression before and after the 18-week endurance training program.

- Alternative Hypothesis (H_1)

There is a significant difference in skeletal muscle gene expression after the 18-week endurance training program.

- Statistical Analysis: Differential Gene Expression Analysis

- Research Question 2
  Which genes are significantly upregulated or downregulated after 18 weeks of endurance training based on P.Value and logFC?
- Null Hypothesis (H_0)
  There are no significantly upregulated or downregulated genes after endurance training.
- Alternative Hypothesis (H_1)
  There are significantly upregulated and/or downregulated genes after endurance training.
- Statistical Analysis:
  Differential gene expression analysis using P.Value < 0.05, with the direction of change determined by logFC.

- Research Question 3

What are the top 20 significantly upregulated genes after 18 weeks of endurance training?

- Null Hypothesis (H_0)

There are no significantly upregulated genes after 18 weeks of endurance training.

- Alternative Hypothesis (H_1)

Several genes are significantly upregulated after 18 weeks of endurance training.

- Statistical Analysis: Genes with positive logFC values were ranked according to P.Value to identify the top 20 significantly upregulated genes after endurance training.

- Research Question 4
  Which differentially expressed genes are the most highly connected within the protein–protein interaction network after endurance training?
- Null Hypothesis (H_0)
  There are no highly connected genes within the protein–protein interaction network of the selected differentially expressed genes.
- Alternative Hypothesis (H_1)
  Some differentially expressed genes show higher connectivity within the protein–protein interaction network.
- Statistical Analysis: STRING Protein–Protein Interaction (PPI) Network Analysis and node degree analysis.
- Research Question 5

Can gene expression profiles distinguish muscle samples before training from those after training?

- Null Hypothesis (H_0)

Gene expression profiles cannot distinguish pre-training samples from post-training samples.

- Alternative Hypothesis (H_1)

Gene expression profiles can clearly distinguish pre-training samples from post-training samples.

- Statistical Analysis: Principal Component Analysis (PCA).

- Research Question 6

Which biological pathways are significantly enriched among the selected differentially expressed genes after endurance training?

- Null Hypothesis (H_0)

There are no significantly enriched biological pathways among the selected differentially expressed genes after endurance training.

- Alternative Hypothesis (H_1)

There are significantly enriched biological pathways among the selected differentially expressed genes after endurance training.

- Statistical Analysis: KEGG Pathway Enrichment Analysis using STRING, with False Discovery Rate (FDR) correction. Pathways with FDR < 0.05 were considered statistically significant.

4. Methods

4.1. Data Preprocessing and Feature engineering

In this phase, raw gene expression data obtained from the GEO2R analysis of dataset GSE111551 was systematically processed to ensure data integrity and prepare it for downstream statistical exploration. Initial data cleaning involved handling unannotated entries, where missing or unassigned gene symbols where it is represented as ---, were replaced with "No Annotation" to maintain structured categorical values across the dataset. Following data hygiene, feature engineering was performed by creating a new categorical variable, Regulated, to classify the expression behavior of each gene. Using a predefined significance threshold of a P-value < 0.05, genes were categorized based on their $\log_2$ fold change (logFC) values into three distinct groups: Upregulated ($\text{logFC} > 0$), Downregulated ($\text{logFC} < 0$), and Not Significant ($P\text{-value} \geq 0.05$). Additionally, feature transformation was applied by calculating $-\log_{10}(P\text{-value})$ to properly scale statistical significance for visualization purposes, such as generating volcano plots. Finally, the preprocessed dataset was exported as Genenome_data_cleaned.xlsx to serve as a standardized input for target identification, functional pathway enrichment, and protein-protein interaction network analysis.

4.2. Statistical Design and models

- Research Question 1: Differential Gene Expression Analysis was used to assess whether skeletal muscle gene expression changed after 18 weeks of endurance training. Statistical significance was evaluated using P.Value, with P.Value < 0.05 used as the significance threshold.
- Research Question 2: Differential Gene Expression Analysis was used to classify significantly altered genes. Genes with P.Value < 0.05 and positive logFC were classified as upregulated, while genes with P.Value < 0.05 and negative logFC were classified as downregulated.
- Research Question 3: Genes with positive logFC were ranked according to P.Value to identify the top 20 significantly upregulated genes after endurance training.
- Research Question 4: STRING Protein–Protein Interaction (PPI) Network Analysis was used to examine interactions among the selected differentially expressed genes. Node degree was used to identify the most highly connected genes within the network.

- Research Question 5: Principal Component Analysis (PCA) was performed using the actual gene-expression matrix to evaluate whether pre-training and post-training samples showed distinct global expression patterns.
- Research Question 6: KEGG Pathway Enrichment Analysis was performed using STRING to identify biological pathways enriched among the selected differentially expressed genes. False Discovery Rate (FDR) correction was applied, and pathways with FDR < 0.05 were considered statistically significant.

5. Findings and Results

- Research Question 1

Does 18 weeks of endurance training significantly change skeletal muscle gene expression?

Based on the differential gene expression analysis of dataset GSE111551, multiple genes showed statistically significant expression changes following 18 weeks of endurance training. Statistical significance was evaluated using P.Value, with P.Value < 0.05 considered significant. Therefore, the results provide evidence that endurance training is associated with changes in skeletal muscle gene expression.

The heatmap was used as a visualization of gene expression patterns among selected genes. It supports the visualization of expression differences but was not used independently to determine statistical significance.

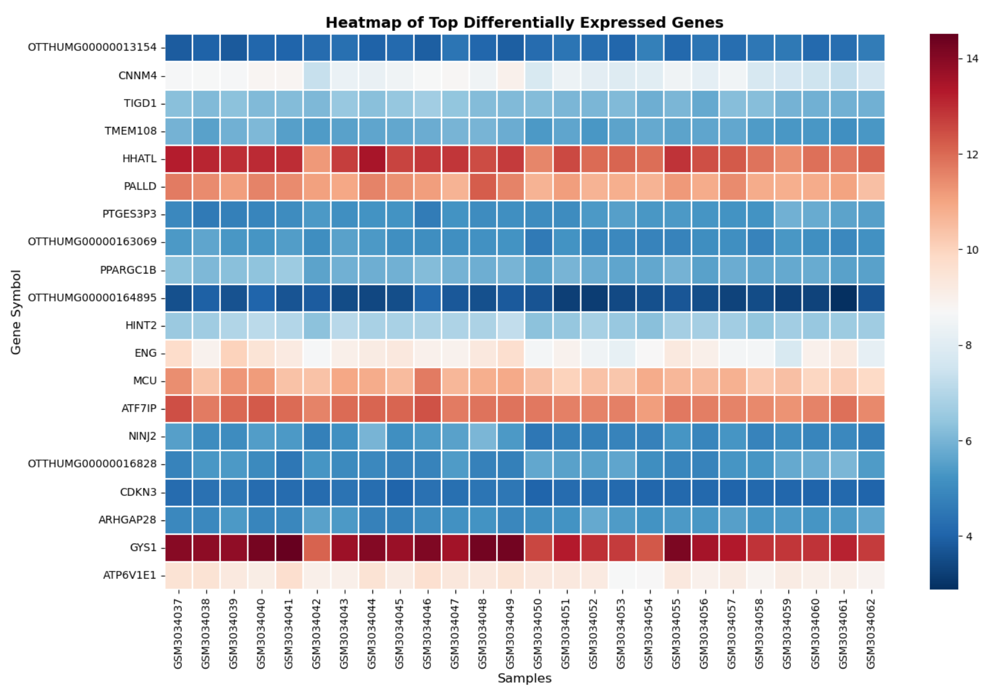


Figure 1: Heatmap for Q1

- Research Question 2

Which genes are significantly upregulated or downregulated after 18 weeks of endurance training based on P.Value and logFC?

Differential gene expression analysis identified both upregulated and downregulated genes following 18 weeks of endurance training. Genes with P.Value < 0.05 and positive logFC values were classified as upregulated, whereas genes with P.Value < 0.05 and negative logFC values were classified as downregulated.

The Volcano Plot illustrates the distribution of gene expression changes according to statistical significance and direction of fold change. Genes located on the positive side of the logFC axis represent upregulated genes, while genes on the negative side represent downregulated genes.

These findings demonstrate that endurance training is associated with bidirectional changes in skeletal muscle gene expression, with some genes showing increased expression and others showing decreased expression after training.

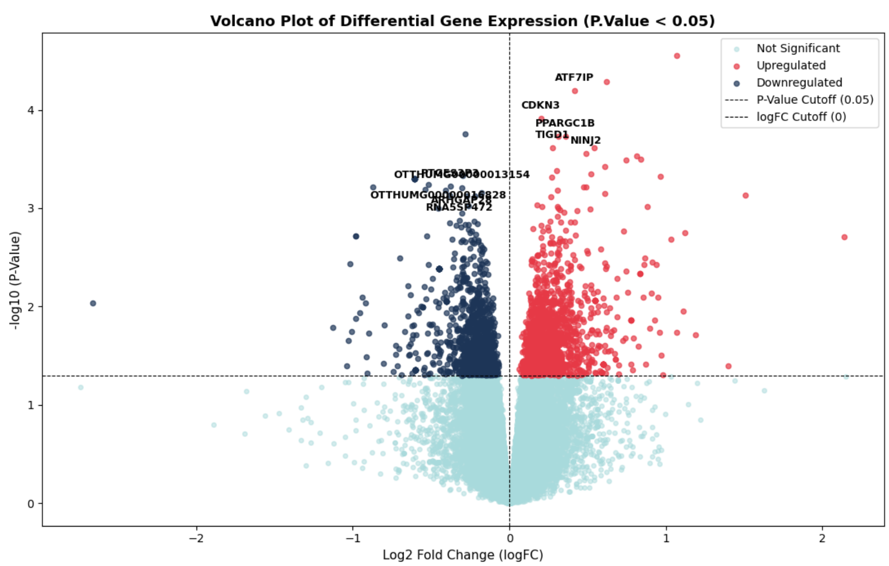


Figure 2: Volcano Plot

- Research Question 3

What are the top differentially expressed genes after endurance training based on statistical significance and fold change?

The top 20 significantly upregulated genes were identified by selecting genes with positive logFC values and ranking them according to statistical significance using P.Value. The resulting genes represent the strongest statistically significant increases in gene expression among the analyzed genes following 18 weeks of endurance training.

The Bar Plot displays the logFC values of the top 20 upregulated genes, allowing comparison of the magnitude of expression increases across the selected genes. These results highlight candidate genes that may contribute to skeletal muscle adaptation following endurance training.

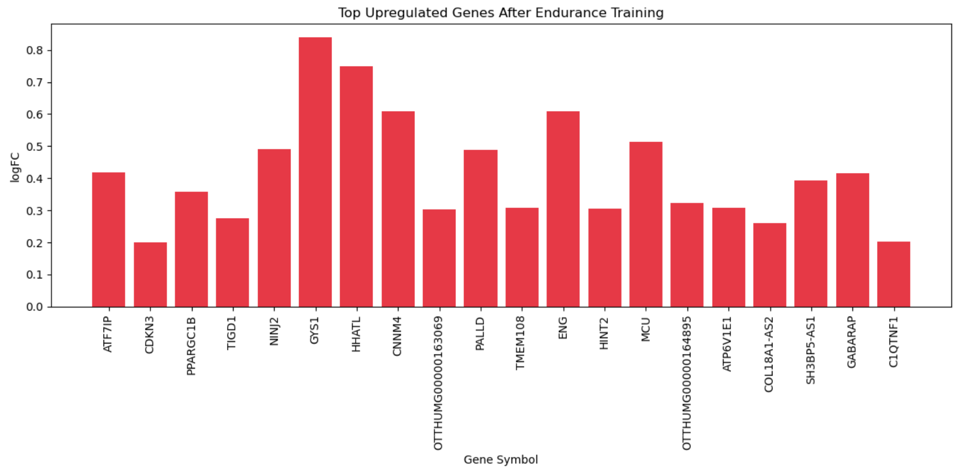


Figure 3: Bar plot

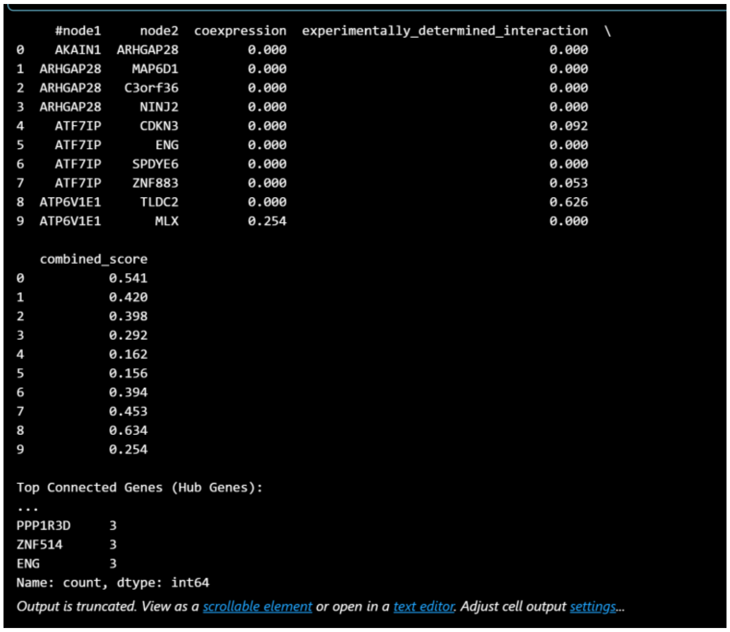
- Research Question 4

Which differentially expressed genes are the most highly connected within the protein–protein interaction network after endurance training?

Protein–protein interaction analysis was performed using STRING to investigate the connectivity among the selected differentially expressed genes. Node degree was used to determine the number of interactions associated with each gene in the network.

ATP6V1E1 showed the highest connectivity with six connections, followed by GYS1 with five connections. ATF7IP, MAIP1, MAP6D1, and ARHGAP28 each showed four connections.

These findings identify ATP6V1E1 and GYS1 as the most highly connected genes within the analyzed protein–protein interaction network. Their higher node degrees indicate that they occupy relatively central positions within the network of selected differentially expressed genes.



```
   #node1     node2  coexpression  experimentally_determined_interaction  \
0   AKAIN1  ARHGAP28         0.000                                  0.000
1 ARHGAP28    MAP6D1         0.000                                  0.000
2 ARHGAP28   C3orf36         0.000                                  0.000
3 ARHGAP28     NINJ2         0.000                                  0.000
4   ATF7IP     CDKN3         0.000                                  0.092
5   ATF7IP       ENG         0.000                                  0.000
6   ATF7IP    SPDYE6         0.000                                  0.000
7   ATF7IP    ZNF883         0.000                                  0.053
8 ATP6V1E1     TLDC2         0.000                                  0.626
9 ATP6V1E1       MLX         0.254                                  0.000

  combined_score
0          0.541
1          0.420
2          0.398
3          0.292
4          0.162
5          0.156
6          0.394
7          0.453
8          0.634
9          0.254

Top Connected Genes (Hub Genes):
...
PPP1R3D     3
ZNF514      3
ENG         3
Name: count, dtype: int64
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
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Figure 4: GO Enrichment

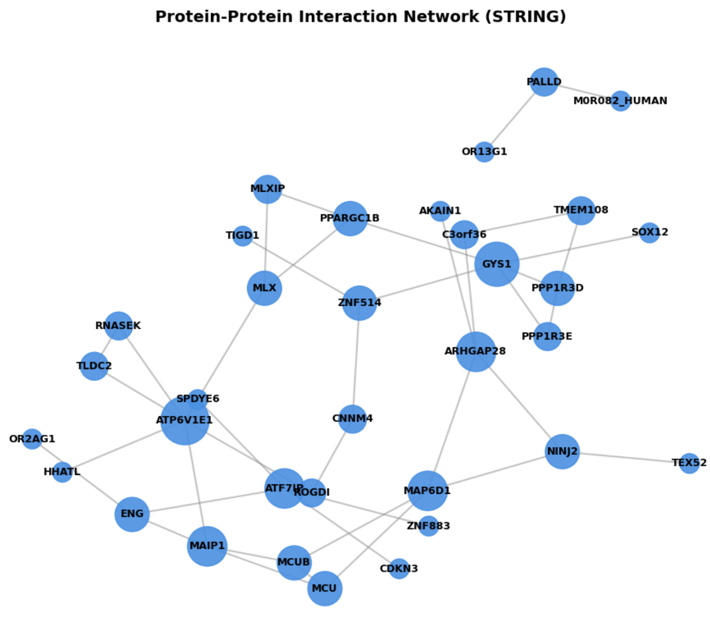

Figure 5: Protein to protein Interaction

- Research Question 5

Can gene expression profiles distinguish muscle samples before training from those after training?

Principal Component Analysis (PCA) did not show clear separation between pre-training and post-training skeletal muscle samples. Considerable overlap was observed between the two groups in the PCA plot.

PC1 explained 17.1% of the total variance, while PC2 explained 9.8%. Together, the first two principal components accounted for approximately 26.9% of the total variance in gene expression.

Therefore, the PCA does not provide evidence of clear global separation between pre-training and post-training gene expression profiles. Although endurance training may be associated with changes in specific genes, these changes were not sufficient to produce distinct clustering of the two conditions in the first two principal components.

The PCA findings are therefore more consistent with the null hypothesis, as clear separation between the pre-training and post-training samples was not observed.

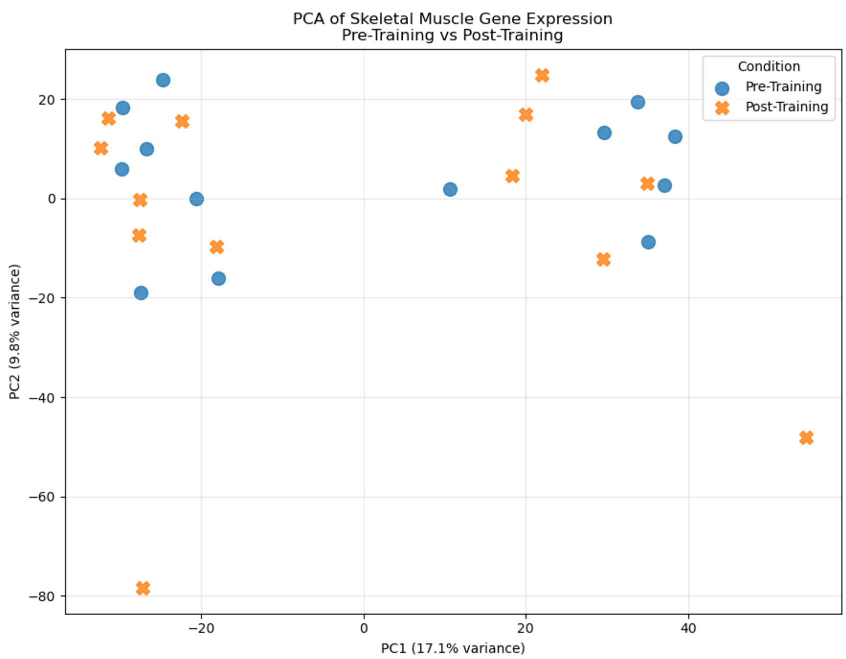

Figure 6: PCA plot

- Research Question 6

Which biological pathways are significantly enriched among the selected differentially expressed genes after endurance training?

KEGG Pathway Enrichment Analysis was performed using STRING to identify biological pathways significantly enriched among the selected differentially expressed genes.

The analysis identified the Insulin Resistance pathway (hsa04931) as significantly enriched. Four genes from the analyzed network were associated with this pathway. The pathway showed a strength of 1.57, a signal of 1.12, and a False Discovery Rate (FDR) of 0.0014.

Since the FDR value was below the predefined significance threshold of 0.05, the Insulin Resistance pathway was considered statistically significant. Therefore, the results provide evidence that at least one biological pathway is significantly enriched among the selected differentially expressed genes.

This finding indicates that genes associated with insulin-related signaling and metabolic regulation are overrepresented among the selected differentially expressed genes. However, enrichment of the Insulin Resistance pathway should not be interpreted as evidence that endurance training causes insulin resistance.

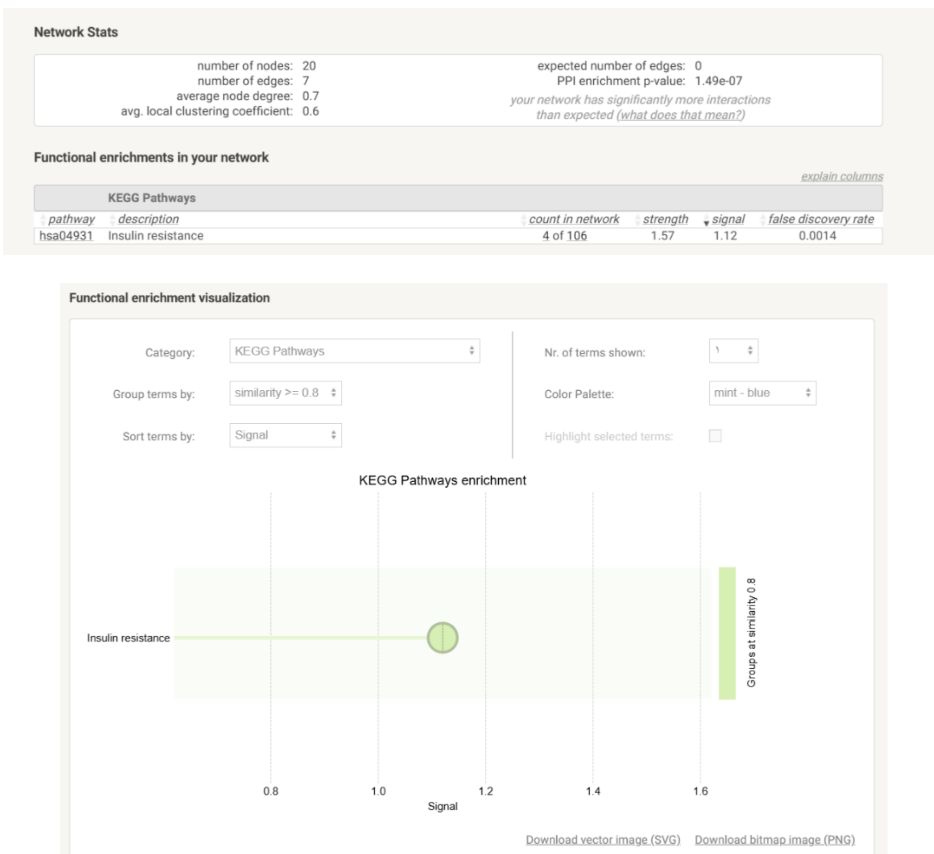

Figure 7: STRING output

6. Limitations and Future improvements

Figure 8: Limitation and Improvements

Category	Aspect / Parameter	Current Limitation in Dataset GSE111551	Recommended Future Improvement
Sample Size & Power	Cohort Size & Diversity	The muscle dataset analyzed in this study included 26 samples (13 pre-training and 13 post-training), which represents a relatively small sample size and may limit statistical power and generalizability	Recruit larger, randomized cohorts including female participants, diverse age groups, and varied baseline fitness levels to improve statistical power and generalizability.
Phenotypic Linkage	Physiological Metadata	Lack of direct sample-level linkage to functional clinical metrics (VO_2peak, individual gains, muscle fiber ratios, or lactate threshold).	Explicitly pair individual transcriptomic profiles with physiological benchmarks, citrate synthase activity, mitochondrial respiration rate) to model training responsiveness.
Resolution	Tissue Heterogeneity	Bulk expression profiling measures average mRNA levels across heterogeneous muscle tissue, masking cell-type-specific responses (Type I vs. Type II fibers, endothelial cells, satellite cells).	Apply single-nucleus RNA sequencing (snRNA-seq) or single-cell transcriptomics (scRNA-seq) to dissect fiber-type-specific adaptations and isolate non-

			muscle cell responses.
Functional Validation	Multi-Omics Integration	Transcriptomic data measures mRNA abundance, which does not always correlate with functional protein levels or post-translational modifications (e.g., PPARGC1B or GYS1 activity).	Adopt a multi-omics approach combining RNA-seq with mass-spectrometry proteomics, phosphoproteomics, and metabolomics to validate functional pathway activity.
Temporal Dynamics	Longitudinal Sampling	Single post-intervention time-point (baseline vs. 18 weeks), which captures chronic adaptation but misses transient transcriptional kinetics.	Implement time-series sampling at multiple training intervals (e.g., acute post-exercise, 2 weeks, 8 weeks, and 18 weeks) to map early vs. late transcriptional dynamics.

7. References

1. GEO2R
2. STRING
3. Jupyter Notebook
4. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111551